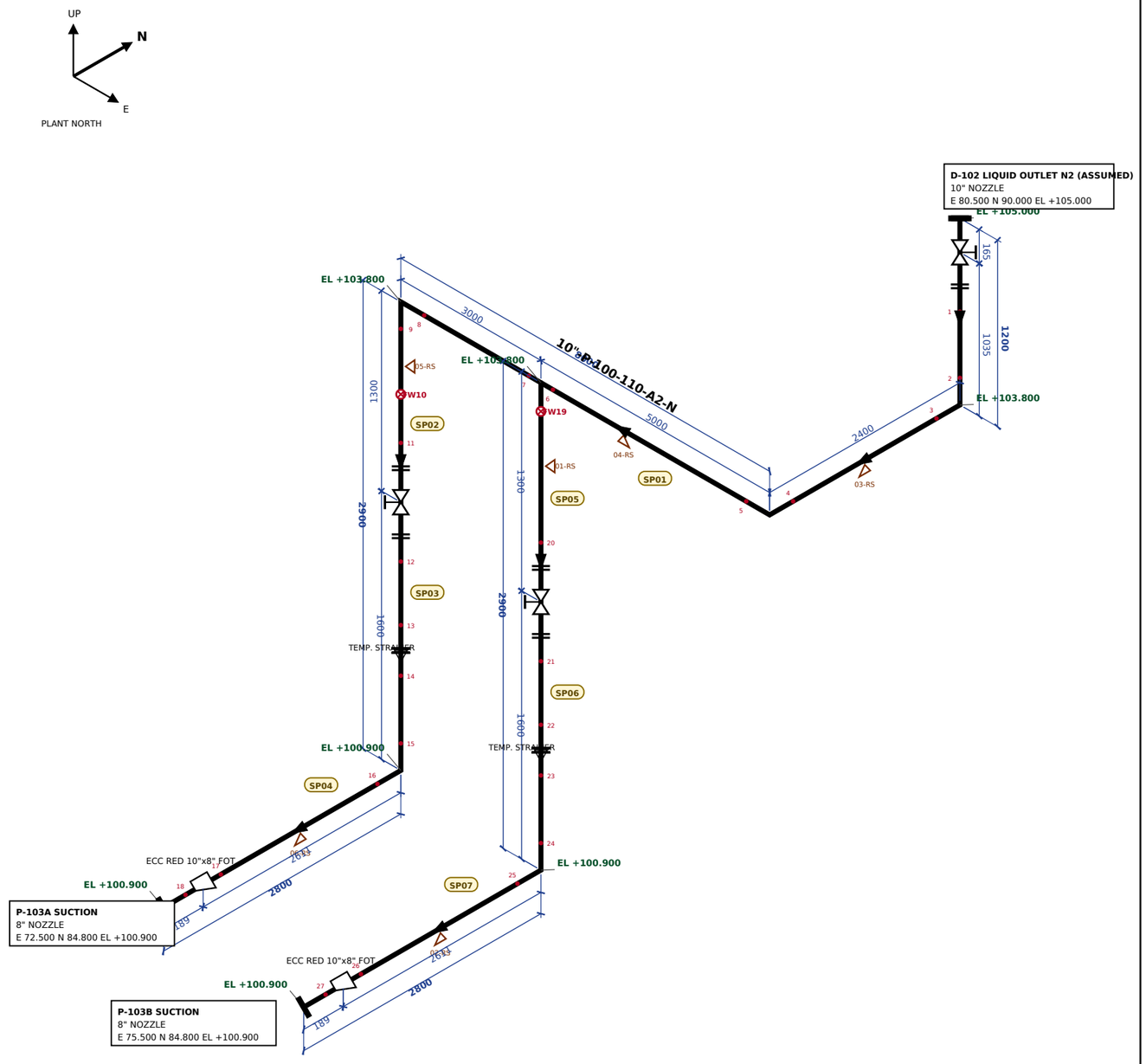
ISOMETRIC VIEW - NOT TO SCALE - DIMENSIONS IN mm, ELEVATIONS / COORDINATES IN m (PLANT GRID)



BILL OF MATERIALS (THIS SHEET)

PT	DESCRIPTION	NPS	QTY
1	FLANGE WN RF CL150 B16.5, bore SCH 40, ASTM A105	10"	9
2	PIPE SCH 40, BE, ASTM A106 Gr.B SMLS HIC/SSC (NACE MR0103)	10"	16.1 m
3	ELBOW 90 LR, BW, SCH 40, ASTM A234 WPB	10"	5
4	TEE EQUAL, BW, SCH 40, ASTM A234 WPB	10"	1
5	REDUCER ECC, BW, SCH 40, ASTM A234 WPB	10" x 8"	2
6	FLANGE WN RF CL150 B16.5, bore SCH 40, ASTM A105	8"	2
7	ERECTION: GATE VALVE OS&Y CL150 RF, A216 WCB, trim 8	10"	3
8	TEMPORARY CONE STRAINER (commissioning)	10"	2
9	GASKET SPW 304/graphite, outer ring, CL150 B16.20	10"	10
10	STUD BOLT SET (nuts) A193 B7 / A194 2H, CL150	10"	10
11	PIPE SUPPORTS (std. dwgs): rest x6	-	6

CUT LENGTHS (mm, ex. weld gap) / SPOOLS

PIECE	NPS	L	PIECE	NPS	L	PIECE	NPS	L
SP01-1	10"	389	SP01-2	10"	1638	SP01-3	10"	4403
SP01-4	10"	2403	SP01-5	10"	250	SP02-1	10"	404
SP03-1	10"	335	SP04-1	10"	419	SP04-2	10"	2141
SP05-1	10"	819	SP06-1	10"	335	SP07-1	10"	419
SP07-2	10"	2141						

SPOOL DEVELOPED LENGTHS (transport limit 12.0 x 3.0 x 3.0 m):
SP01=11.9m SP02=0.5m SP03=0.5m SP04=3.7m SP05=0.9m SP06=0.5m SP07=3.7m

NOTES

1. Pipe/fittings/flanges per class A2 (CFU-000-PI-SPC-001); BW fittings ASME B16.9, flanges B16.5/B16.47A RF.
2. Hydrotest 5.2 barg per ASME B31.3 345.4 (vents at high points, drains at low points).
3. PWHT: YES - service (sour/HIC) 595-650 C. NDE: 20 % RT of BW; HB <= 200.
4. Supports: RS rest/shoe, GD guide, ANC anchor, SPR variable spring - tags PS-<area><seq>-nn.
5. Welds: dot = shop weld, crossed circle FW = field weld; spools <= 12 m for transport.
6. Pump suction: no pockets, header falls to pumps; eccentric reducers 10"x8" FLAT ON TOP; temporary straine
7. Pump nozzle 8" assumed (API 610 vendor data pending).
8. Vortex breaker in D-102 outlet nozzle (by vessel).
9. Common header, equal branch to each pump.

LINE NUMBER	FROM	TO	PIPING CLASS
10"-P-100-110-A2-N	D-102	P-103A/B	A2 - 150# CS ASTM A106 Gr.B HIC-resistan
DESIGN P / T	OPER. P / T	TEST P (HYDRO)	PIPE WALL
3.5 barg / 75 C	0.69 barg / 45.0 C	5.2 barg (1.5xPxST/S=1.0)	SCH 40 (9.27 mm)
INSULATION	PWHT	NDE	STRESS
N 0 mm	YES - service (sour/HIC) 595-650 C	20 % RT of BW; HB <= 200	CAT 2 - T > 150 C or equipment-connected
P&ID	FLUID / PHASE	PAINT	SPOOLS / WELDS
CFU-100-PR-PID-009	P / L Reflux pump suction	Epoxy/PU system	7 spools; 25 shop / 2 field

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)

GENERIC REFINERY - GULF COAST
100 KBPSD CRUDE & VACUUM DISTILLATION UNIT

PIPING ISOMETRIC 10"-P-100-110-A2-N

D-102 TO P-103A/B SUCTION

DISCIPLINE	STAGE	SCALE	SHEET
PIPING	FEED	NTS	1 OF 1

DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02

DRAWING NUMBER: **CFU-100-PI-ISO-010**

REV: **A**